

SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

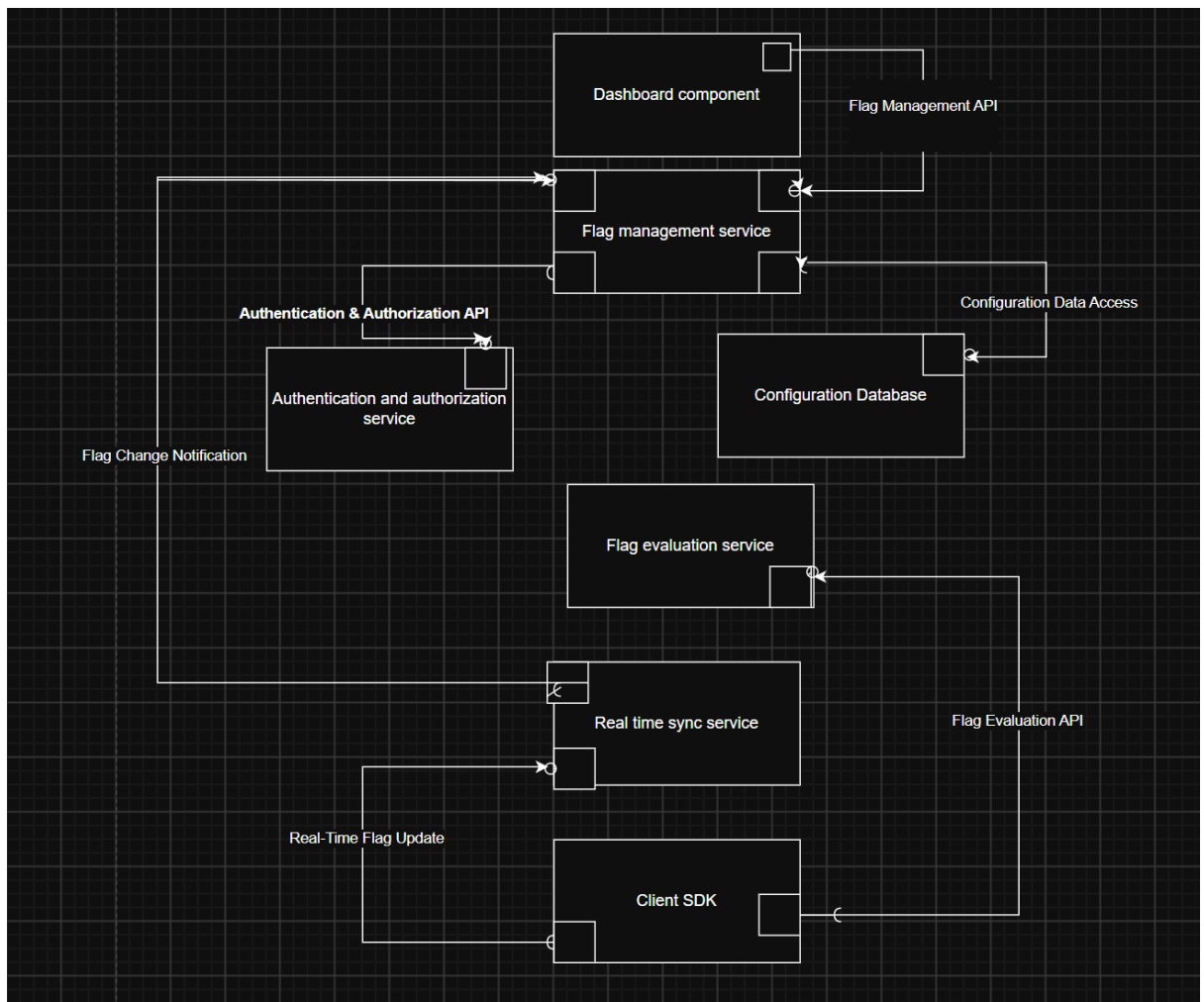
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CSE-D

Feature flag and dynamic config manager

1) Component Diagram



2) Architecture Selection

I chose **Microservices Architecture** for the **Feature Flag & Dynamic Config Manager System**.

Reason 1 — Independent scaling

My system has a **feature-flag evaluation function** that is used by client applications and has a strict **<15 ms performance requirement**.

With microservices, the evaluation functionality can be separated from less frequently used functions such as flag configuration or administration and scaled independently.

Reason 2 — Fault isolation

My system performs several different functions:

- Flag management
- Rollout/targeting configuration
- Flag evaluation
- Environment configuration
- Real-time synchronization
- Authentication/auditing

Microservices allow these functions to be separated into services, providing **fault isolation**.

Security Advantage

The Authentication & Authorization Service can be separated from the other services and used to control access to operations that change feature flags. This supports the system's requirement for authentication and role-based permissions, ensuring that only authorized Software Engineers and Release Managers can make configuration changes.

Performance Benefit

The Flag Evaluation Service can be independently scaled to handle a high number of feature-flag evaluation requests. Separating it from configuration and administrative operations helps it meet the system's strict requirement of completing feature-flag evaluations in under 15 ms.